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Clustering and Annotation of Cell Types in DLBCL via Spatial Transcriptomics with

by

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Abstract

Diffuse large B cell lymphoma (DLBCL) is highly heterogeneous, complicating both classification and treatment. Using CosMx Spatial Molecular Imaging at single-cell resolution, we profiled four fields of view comprising 11,382 cells and 989 genes from a diagnostic DLBCL specimen. After quality control and dimensionality reduction, hierarchical clustering resolved six transcriptionally coherent groups spanning the major tissue compartments. Five clusters were biologically interpretable: activated B cells, a B cell to plasmablast and plasma continuum, fibroblasts with extracellular matrix programmes, macrophages and monocytes, and secretory epithelial cells. These identities were validated by consistent marker sets across differential expression methods and confirmed against curated reference databases. A sixth cluster localised to tissue edges and displayed low-content signatures; it was retained for completeness but excluded from biological interpretation. Overall, the results show that hierarchical clustering captures meaningful nested organisation in spatially resolved DLBCL data and provides a reproducible framework for cell-type annotation that can be extended to further samples to inform future biological and therapeutic work.

Introduction

Diffuse large B-cell lymphoma (DLBCL) is the most common aggressive type of non-Hodgkin lymphoma, comprising a significant proportion of adult lymphoma cases globally^{1,2}. Despite advances in chemo-immunotherapy, a considerable fraction of patients fail to achieve long-term remission, underscoring the disease's biological and clinical heterogeneity^{3,4}. This heterogeneity arises not only from the malignant B cells but also from the complex tumour microenvironment (TME), which includes various immune and stromal

populations affecting survival, progression, and therapeutic outcomes⁵. Traditional methods such as bulk transcriptomics and immunohistochemistry have elucidated genetic drivers and broad cell-of-origin classifications, yet they often obscure cell-level heterogeneity or lack cellular resolution^{6–8}. Single-cell RNA sequencing (scRNA-seq) has provided higher resolution insights into cell subpopulations within DLBCL^{9,10}, although it lacks spatial context crucial for understanding lymphoid architecture and tumour-immune interactions¹¹. Spatial transcriptomics techniques, including imaging-based platforms like the CosMx Spatial Molecular Imager, now enable mapping of gene expression at single-cell resolution with preserved tissue context, facilitating finer characterization of cellular heterogeneity in DLBCL¹². Central to analysis is clustering, which assigns cells into putative groups based on expression profiles. Most clustering approaches yield flat partitions, with independent, unconnected clusters; however, biological organization is often hierarchical, where broad immune or stromal compartments subdivide into functionally distinct subsets^{13,14}. Capturing this layered structure is particularly important in heterogeneous diseases like DLBCL, where malignant, immune, and stromal programs intersect^{15–17}. This study centres on hierarchical clustering to represent the multi-level organization of DLBCL tissues in spatial transcriptomics datasets generated by CosMx. To progress beyond computational grouping, clusters are annotated and biologically interpreted using differential gene expression and marker-based feature selection^{18,19}. This framework aims to clarify DLBCL tissue composition and organization at single-cell resolution, linking malignant heterogeneity with TME features^{20,21,22}. Methodologically, the work evaluates hierarchical clustering's advantage over flat clustering for spatial and single-cell data. Biologically, it contributes to understanding lymphoma complexity and supports precision oncology by integrating robust computational methods with biological annotation^{20,23,22,24}. Together, these advances deepen

insights into DLBCL heterogeneity and foster progress toward personalized therapeutic strategies.

Hypothesis and Aims

Hypothesis

We hypothesise that hierarchical clustering can reveal meaningful hierarchical relationships among malignant, stromal, and immune cell populations in Diffuse Large B-Cell Lymphoma (DLBCL) using CosMx Spatial Molecular Imager data. By applying distance- and linkage-based variations of hierarchical clustering, and integrating gene expression profiles with established marker references, we expect to identify both major cellular compartments and potential subpopulations. These clusters, once annotated against curated reference databases and atlases, should capture the heterogeneity of the tumour microenvironment more effectively than flat clustering approaches.

Aims

1. To evaluate hierarchical clustering strategies systematically compare alternative distance metrics (e.g. Euclidean, correlation, cosine) and linkage methods (e.g. average, complete, Ward's) for grouping CosMx single-cell expression data, assessing their impact on cluster granularity and biological interpretability.
2. To identify distinct subpopulations within DLBCL determine whether hierarchical clustering can resolve fine-grained subsets within malignant B cells, macrophages, and T-cell compartments, thereby reflecting the cellular diversity observed in recent single-cell atlases of DLBCL.

3. To integrate differential expression analysis with annotation – cross-validate top cluster-specific markers against reference resources (e.g. ACT, CellMarker 2.0, CellKb, and published DLBCL single-cell studies) to assign biologically meaningful labels to clusters.

Methods

Data source and specimen

The study analysed a diagnostic diffuse large B-cell lymphoma (DLBCL) specimen sourced from the CCLG Vivo Biobank. Tissue processing for RNA in situ profiling was performed by Dr Matthew Pugh at the University of Birmingham. Imaging and molecular readout were carried out on the CosMx Spatial Molecular Imager at Birmingham Tissue Analytics, University of Birmingham. Although CosMx generates spatially resolved measurements, only the gene expression matrices were used for the present work; spatial coordinates were not used in any downstream analysis. The raw data were provided as a Seurat RDS file (sample_1_Anne_FOV103_106.rds) containing four fields of view. Per-FOV cell counts were: FOV 103, 2,806 cells; FOV 104, 2,865 cells; FOV 105, 2,656 cells; FOV 106, 3,077 cells, giving a total of 11,404 cells. The targeted transcript panel comprised 989 genes. The RDS object was read in R and exported to plain-text comma-separated values (CSV) for analysis in Python, preserving one matrix of cells by genes and a companion cell-level metadata table. All downstream steps refer to these CSV exports.

Data Processing and Quality Control

The bioinformatic analysis was primarily conducted using Python, leveraging the scanpy library for single-cell data manipulation¹⁹. Supporting libraries included pandas for data

structuring and matplotlib and seaborn for visualization. The initial step involved creating an AnnData object to efficiently store the single-cell expression matrix and associated metadata. A rigorous quality control (QC) process was applied to ensure the integrity of the data¹⁹. The raw dataset, as initially loaded, comprised 11,404 cells and 989 genes. We began by visualizing the raw data, and a key observation was the distinct extended tail in the distribution of total counts per cell and genes detected per cell. This tail is a clear indicator of low-quality cells, which are characterized by a low number of detected genes. To address this, we implemented a series of filtering steps.

First, we removed cells with an excessively high mitochondrial gene content, setting a threshold of 15%; however, no cells were removed at this stage, indicating a low level of mitochondrial contamination in the dataset. Next, a total counts filter was applied, removing cells with a total count less than or equal to 1,000, which resulted in the removal of 12 low-quality cells, leaving 11,392 cells. A subsequent filter on the number of genes detected per cell (`n_genes_by_counts`) removed cells with fewer than 350 detected genes, resulting in 11,382 cells. Finally, a gene-level filter was applied to remove genes expressed in fewer than three cells, but this step did not remove any genes from the dataset.

Following these comprehensive QC steps, the dataset was reduced to a final count of 11,382 cells and 989 genes for subsequent analysis.

Normalisation and transformation

After quality control, we performed library-size normalization using `sc.pp.normalize_total` to equalize sequencing depth across cells, followed by log transformation (`sc.pp.log1p`). The normalized data were used for preliminary dimensionality reduction and visualization on the HVG matrix, whereas `log1p`-transformed expression values without z-scoring were retained for all downstream inferential analyses, including cluster-wise `log2` fold-change estimation,

random forest feature importance assessment, and MAST differential expression testing^{25,26}.

This approach preserves biological interpretability of effect sizes on the original measurement scale, which is particularly important in a targeted-panel context where inappropriate scaling could distort fold-change estimates and confound biological conclusions.

Highly variable genes Selection

Because the CosMx panel comprised 989 pre-selected genes for lymphoid tissue, we applied a percentile-based approach to identify highly variable genes that adapts to the dynamic range of a targeted panel^{27,28}. For each gene, mean expression and normalized dispersion were calculated across all 11,382 cells. Genes were flagged as HVG candidates only if both statistics exceeded the 75th percentile of their respective distributions, thereby retaining genes that were simultaneously well expressed and variable. These percentile cut-offs avoided brittle absolute thresholds and ensured that variability was judged relative to the behavior of the full panel. Although this HVG list was compiled for interpretability checks, all 989 genes were still retained for PCA and clustering so that curated markers just below the top quartile were not inadvertently discarded. For differential testing and annotation, we further required that a gene be detected in at least 25% of cells within the cluster being tested, a prevalence rule that limits spurious markers driven by rare cells while aligning with practice in lymphoid single-cell studies. This approach follows standard methodology, where data scaling is applied after highly variable gene selection to ensure robust dimensionality reduction and clustering, while preserving biological interpretability in marker detection analyses

Dimensionality reduction

After HVG selection, the data was scaled then PCA was run on the scaled matrix. An elbow plot was generated to assess explained variance per principal component, revealing

diminishing returns beyond the first several components. We selected 12 principal components for downstream analysis, preserving sufficient biological variation while excluding noise components. This choice of 12 PCs balanced capturing meaningful structure while avoiding the inclusion of technical noise. The 12-PC embedding was used for all subsequent steps including hierarchical clustering with Ward's method, validation metrics (silhouette, Davies-Bouldin, and Calinski-Harabasz indices), and differential expression analyses

Hierarchical clustering and choice of k

Clustering was performed in the twelve-dimensional PCA space derived from the HVG scaled matrix using agglomerative hierarchical clustering with Ward's linkage on Euclidean distances^{19,29}. Ward's criterion was selected because it directly minimizes within-cluster variance, which is appropriate when using continuous, centered expression features, and it yields dendrograms that are straightforward to interpret biologically for hierarchical relationships among immune cell states. We cut the tree at six clusters, $k = 6$, because this cut height maximized the Calinski-Harabasz index and minimized the Davies-Bouldin index while giving a silhouette score that, although modest, was stable³⁰⁻³².

Differential expression and marker discovery

All differential analyses were performed in a one-versus-rest design for each cluster to identify cluster-enriched marker genes. Tests used the unscaled log1p matrix for all 11,382 cells $\times$ 989 genes, with a gene considered only if it was detected in $\geq 25\%$ of cells within the index cluster (prevalence rule). For every method we also reported an effect size as a cluster-level log2 fold change (log2FC), defined as the difference between the mean $\log_2(1 + \text{expression})$ in the index cluster and the mean $\log_2(1 + \text{expression})$ in the remaining cells; this keeps ranks comparable across methods and aligns with the targeted-panel context.

This wording harmonizes the gene universe (all 989 genes) with the prevalence rule stated elsewhere.

Wilcoxon rank-sum test

For each cluster, we asked which genes were upregulated in that cluster relative to all other cells. The Wilcoxon rank-sum test characterizes distributional shifts in a non-parametric way: pooled cell-level expression values are ranked, and the test asks whether ranks from the index cluster tend to be higher than those from the rest. This test does not assume normality and is robust to outliers, which suits CosMx targeted panels where many genes show zero-inflated, skewed distributions³³.

Log2 fold change was explicitly defined as:

$$\log_2 \text{FC} = \log_2 (X_{\text{cluster}} + 1) - \log_2 (X_{\text{rest}} + 1),$$

where X_{cluster} is the mean expression within the index cluster and X_{rest} is the mean across all other cells.

We ranked genes by effect size first (largest positive log2 fold change) and used adjusted P-values as a tiebreaker, to prioritize markers that were both statistically convincing and biologically large in effect. The top twenty Wilcoxon markers per cluster were defined as those with the highest positive log2 fold change that also passed the 25% prevalence rule.

Random forest classifier

To capture multigene, possibly non-linear signal, we trained a stratified 5-fold cross-validated random forest to predict cluster labels from all genes³⁴. In each fold ~80% of cells formed the training set and ~20% formed the test set, preserving class proportions. After confirming stable accuracy across folds, we refit the model on all cells to extract feature importances

(mean decrease in impurity) and computed class-specific contributions so we could score genes for each cluster against the rest. To align directionality with biology, we considered only genes whose average expression was higher inside the cluster than outside, then ranked by class-specific importance and took the top twenty per cluster. This approach is appropriate here because combinations of moderately expressed transcripts (common in immune activation programs) can be highly discriminative even when no single gene is extreme on its own.

MAST hurdle modelling

MAST was run in a one-versus-rest design for each cluster using its two-part (“hurdle”) model²⁶. This framework assumes that expression can be decomposed into (i) a detection component (logistic regression, testing whether a gene is on in a greater fraction of cells in the index cluster) and (ii) a continuous component (Gaussian model, testing whether expression levels among detected cells are higher in the index cluster). These are combined into an overall likelihood-ratio statistic:

$$p_hurdle = f(p_det, p_cont),$$

where a strong effect in either part drives p_hurdle to small values.

Log2 fold change was defined as the difference in mean log2 expression between the cluster and the rest (as above). Genes were retained if they showed FDR-adjusted $p_hurdle < 0.05$ and a positive log2 fold change. From this set of significant, up-regulated genes, the top twenty per cluster were chosen by effect size, alongside the Wilcoxon and random-forest results.

We selected Wilcoxon, random forest, and MAST as complementary approaches because each captures different aspects of gene cluster separation in CosMx data. Wilcoxon is robust

to non-normal distributions and highlights consistent marginal shifts. Random forest captures multigene, non-linear structure that often underlies immune states. MAST accounts for zero inflation by separating detection from abundance, matching the on/off biology of immune activation and graded expression levels. In practice, Wilcoxon surfaced canonical markers with strong marginal shifts; random forest prioritized multigene combinations that best separated each cluster; and MAST flagged genes where biology manifested either as higher detection, higher expression among positives, or both.

After selecting the Top 20 genes per cluster from each method, we quantified overlap between the three lists. By design, Wilcoxon, random forest, and MAST each contributed 20 genes per cluster. Cluster 1 showed no triple-overlap (0 genes common to all three lists). The remaining clusters showed the following numbers of triple-overlap genes: Cluster 2 = 7, Cluster 3 = 6, Cluster 4 = 18, Cluster 5 = 17, and Cluster 6 = 12. These counts confirm that the methods are complementary some clusters (e.g., 4–5) exhibit strong agreement across tests.

Cell-type annotation and spatial validation

After deriving the Top 20 markers per cluster from each method (Wilcoxon, Random Forest, MAST), we annotated cell identities by querying three reference databases ACT³⁵, CellMarker 2.0³⁶, and CellKb³⁷ using each cluster's marker set. We then finalized labels against the CZ Cellxgene Gene Expression portal³⁸, and wrote the consensus cell types onto the UMAP for visual verification. As a quality check, we examined cross-method agreement and found no triple-overlap for Cluster 1, prompting a spatial review across all FOVs: Cluster 1 cells were concentrated at tissue edges and showed low-content patterns, indicating mixed quality²⁴. This suggests Cluster 1 is heterogeneous a mixture of sub-populations and

edge/low-quality cells so different DE tests (Wilcoxon, MAST, RF) surfaced different weak markers, eliminating intersections and explaining the absence of overlap.

Results

Data Pre-processing and Quality Control

We began with 11,404 cells and 989 genes. A mitochondrial filter was applied to remove cells in which more than 15% of transcripts mapped to mitochondrial genes, as such cells are often stressed or dying; in practice, none of the cells exceeded this threshold, so the totals were unchanged. We then applied a gene-level filter, discarding genes expressed in fewer than three cells, which helps reduce noise from very sparse features; again, this did not alter the counts. Next, a total-counts filter was used to exclude cells with fewer than 1,000 detected transcripts, since very low library sizes usually reflect poor capture efficiency, and this removed 12 cells. Finally, a gene-detection filter excluded cells with fewer than 350 detected genes, reflecting low-content profiles unlikely to support reliable clustering; this step removed a further 10 cells. After all filters, 11,382 cells and 989 genes were retained for downstream analysis.

	Step	Number of Cells	Number of genes
1	Raw dataset (as loaded)	11,404	989
2	After mitochondrial filtering (<15%)	11,404	989
3	After gene-level filter (min_cells $\geq$ 3)	11,404	989
4	After total counts filter ($\leq$ 1,000)	11,392	989
5	After gene detection filter (n_genes_by_counts $\leq$ 350)	11,382	989

Table 1. The table shows the number of cells and genes retained after each quality control filter. Cells with high mitochondrial transcript fractions ($>15\%$) were flagged as potentially stressed or dying but none were removed. Genes expressed in fewer than three cells were discarded, although this did not alter the totals. Cells with fewer than 1,000 detected transcripts were removed as low-quality profiles (12 cells), and an additional 10 cells with fewer than 350 detected genes were excluded as low-content profiles. After all filters, 11,382 cells and 989 genes remained for downstream analysis

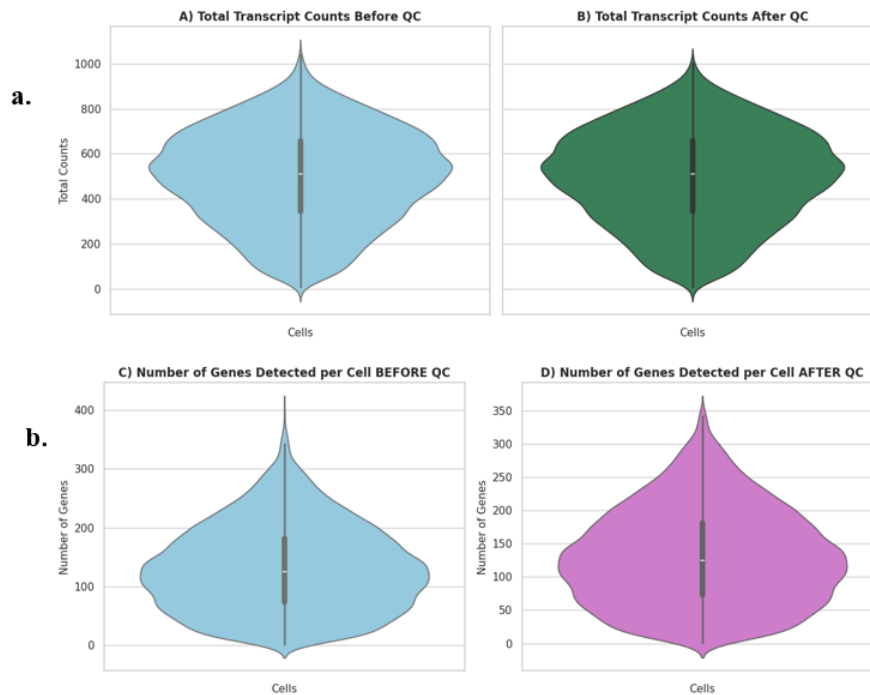


Figure 1. (a) Total transcript counts per cell before (A) and after (B) QC. Violins show the full distribution (width $\propto$ density). The central box marks the interquartile range (IQR, 25th–75th percentiles), and the horizontal line indicates the median. QC removes low-count outliers and tightens the distribution while preserving the upper range expected for intact cells. (b) Number of genes detected per cell before (C) and after (D) QC. Post-QC, the distribution shows a higher median and a reduced lower tail, consistent with the removal of low-complexity cells.

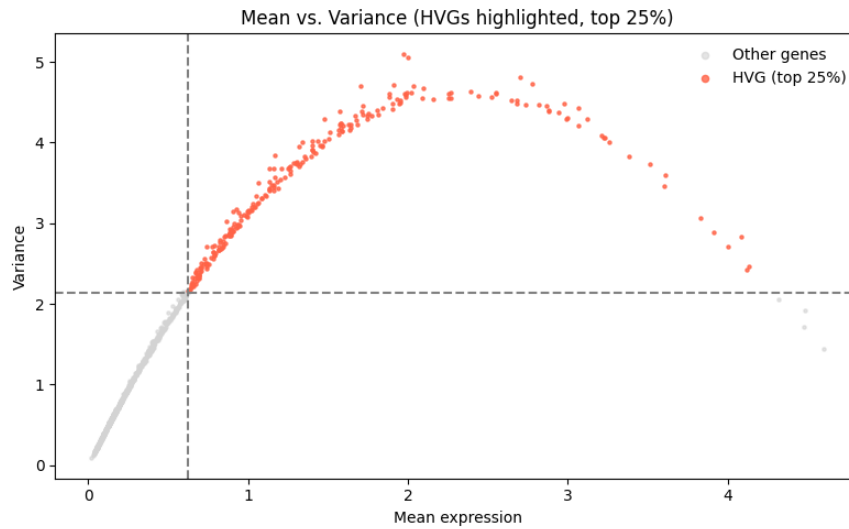


Figure 2. Mean–variance relationship and HVG selection. Scatter of per-gene mean expression versus variance computed on log1p-normalised counts. Orange points mark the highly variable genes (HVGs) defined as those above the 75th percentile for both mean expression and dispersion ($n = 244$). These HVGs were retained for downstream PCA, UMAP, and hierarchical clustering; grey points are all other genes. The dashed vertical and horizontal lines indicate the 75th-percentile cut-offs used to define HVGs. The curved trend reflects the expected mean–variance dependence in single-cell data

Dimensionality reduction and hierarchical clustering

We evaluated the Ward partition in the 12-PC Euclidean space using three standard indices. The silhouette coefficient (range -1 to 1 ; higher is better) averaged 0.163 , indicating partial overlap and/or unequal cluster sizes. The Davies–Bouldin index (≥ 0 ; lower is better) was 1.654 , consistent with relatively compact, separated clusters. The Calinski–Harabasz index (≥ 0 ; unbounded above; higher is better) was 2384.693 , supporting good between-cluster separation relative to within-cluster dispersion. Taken together, the low-to-modest silhouette

alongside favourable Davies–Bouldin and Calinski–Harabasz values supported $k = 6$, which we retained for downstream interpretation.

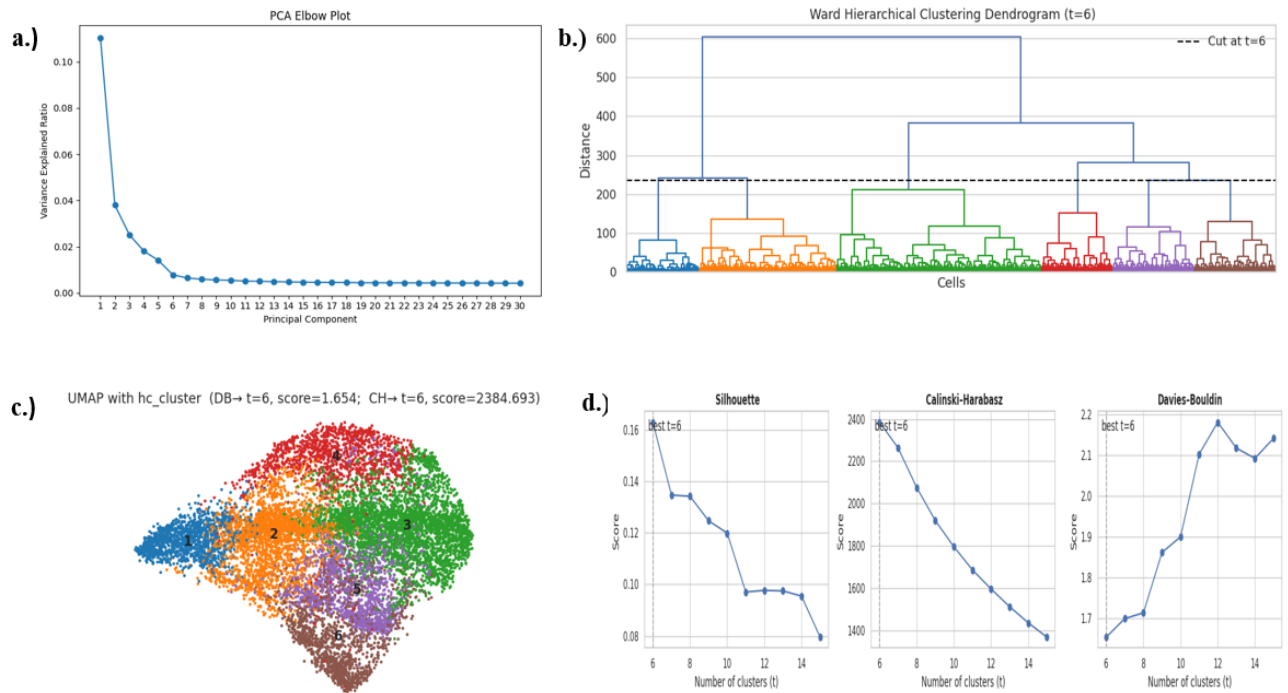


Figure 3. (a) PCA elbow plot on the HVG scaled set. The curve drops steeply then plateaus, indicating diminishing returns beyond the leading components; 12 PCs were retained for neighbour-graph construction, UMAP and hierarchical clustering. (b) Ward dendrogram computed in the 12-PC space. The horizontal dashed line marks the cut that yields six clusters ($k=6$); branch colours denote membership below the cut. (c) UMAP embedding coloured by cluster IDs 1–6 from the Ward solution. Partially overlapping yet distinct neighbourhoods are evident, as expected for closely related immune/stromal compartments. Internal indices for this partition are Davies–Bouldin = 1.654 and Calinski–Harabasz = 2384.693; the corresponding silhouette is (d) Internal validity across candidate k values: Silhouette (higher is better) peaks at $k=6$ then declines; Calinski–Harabasz (higher is better) is maximal at $k=6$; Davies–Bouldin (lower is better) is minimised near $k=6$. Together this support selecting six clusters for downstream interpretation despite the modest silhouette.

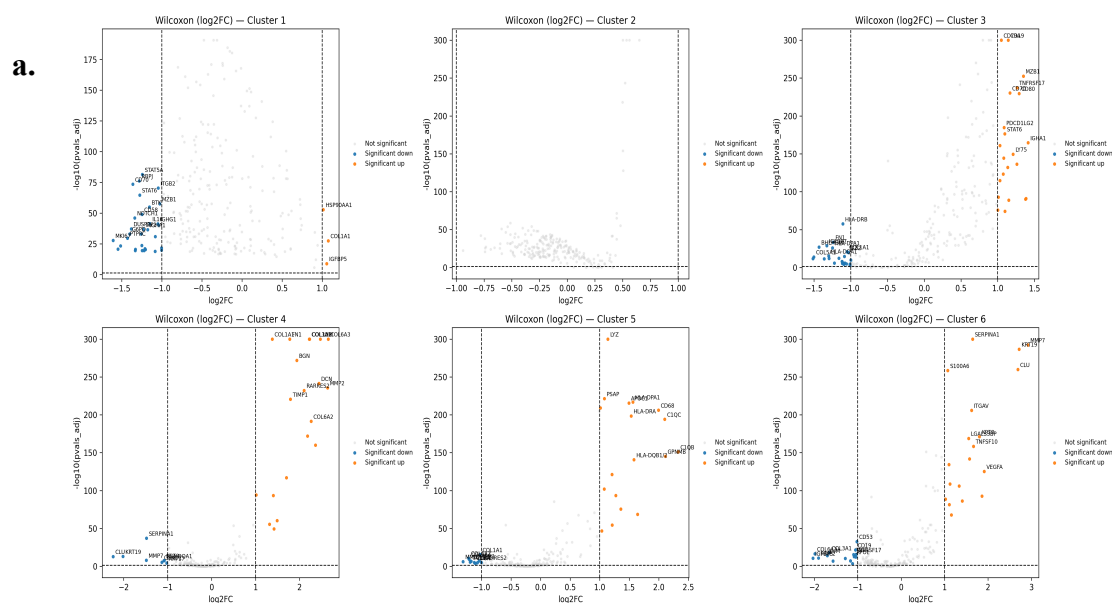
Differential expression and marker discovery

Marker detection was performed one-vs-rest on the log1p-normalised, unscaled HVG set.

Three complementary approaches were used, and for each cluster we extracted a Top 20 list per method:

- Wilcoxon rank-sum (BH-FDR): genes ranked by smallest FDR; ties broken by log2FC. (The displays show effect direction; for interpretation we prioritized genes with positive log2FC in the target cluster.)
- Random-Forest (RF) feature importance: one-vs-rest classifier; fold-averaged, z-scored importances. The Top 20 comprise the highest-importance genes with positive effect (higher mean expression in the target cluster).
- MAST-like hurdle model: zero-inflation aware; BH-adjusted p with log2FC as effect size. The Top 20 are FDR-significant, positively enriched genes with the largest log2FC.

Across clusters 2–6 there was clear overlap of markers across all three methods. Cluster 1 showed no three-way consensus.



b.

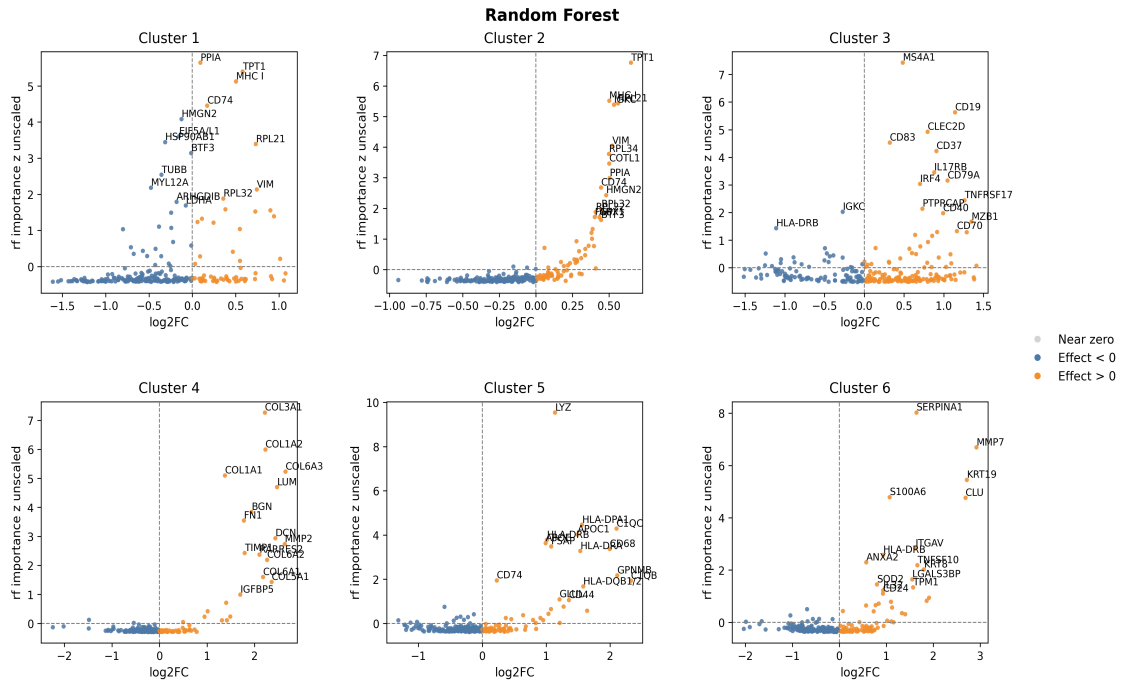


Figure 4: (a) Wilcoxon one-versus-rest volcano plots for clusters 1–6, showing $\log_2$ fold change (x-axis) against $-\log_{10}(\text{FDR-adjusted P-value})$ (y-axis). Dashed lines mark $\log_2\text{FC} = 1$ and $\text{FDR} = 0.05$. Genes passing both thresholds are highlighted (orange, up-regulated; blue, down-regulated; grey, not significant). Labels denote exemplar markers per cluster. In some cases, strongly negative $\log_2\text{FC}$ values did not correspond to high significance. This reflects the prevalence rule ($\geq 25\%$ detection within the cluster) and the Wilcoxon test properties: genes expressed in few cells can yield large fold changes but modest statistical support, an expected behaviour in targeted CosMx panels with zero-inflated distributions. First, upregulated genes ($\log_2\text{FC} > 0$) were sorted by p-value and $\log_2\text{FC}$. Since there weren't enough genes meeting the strict visualization thresholds on the volcano plot, the algorithm still selected the "best" genes by their relative rankings. The genes selected had smaller effect sizes and/or higher p-values compared to the thresholds used for the volcano plot coloring. (b) Random forest profiles for clusters 1–6, plotting $\log_2\text{FC}$ (x-axis) against feature importance (y-axis). Points are coloured by effect sign as in panel (a). Labels highlight the most influential positively enriched genes per cluster.

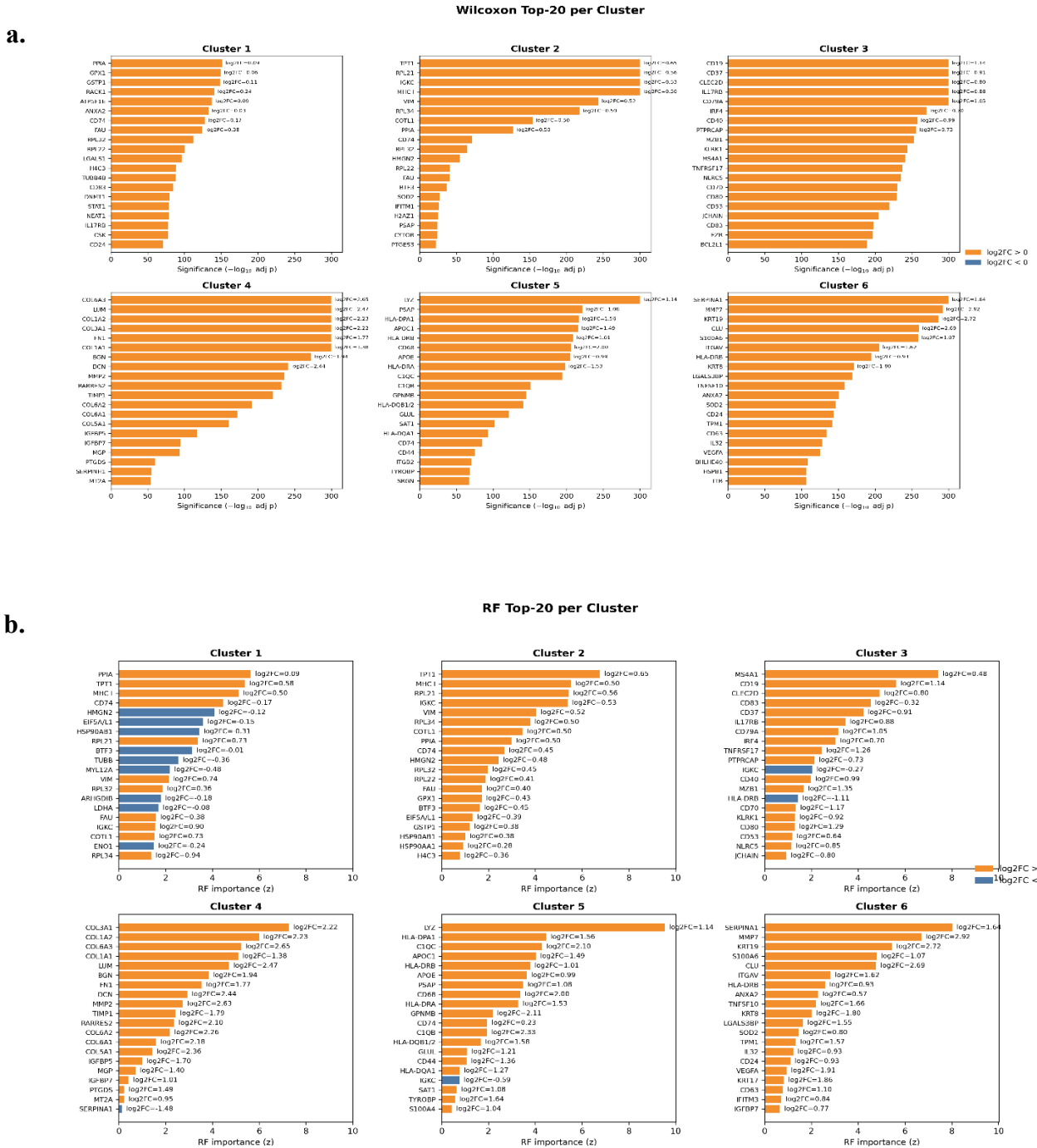


Figure 5: (a) Wilcoxon Top 20 bar charts per cluster, ordered by $-\log_{10}(\text{FDR})$; bar colour indicates effect (orange: $\log_2\text{FC} > 0$; blue: $\log_2\text{FC} < 0$). (b) Random-Forest Top 20 bar charts per cluster, ordered by RF importance (z); bars coloured by effect as above. Only positively enriched genes were retained for RF Top 20 display.

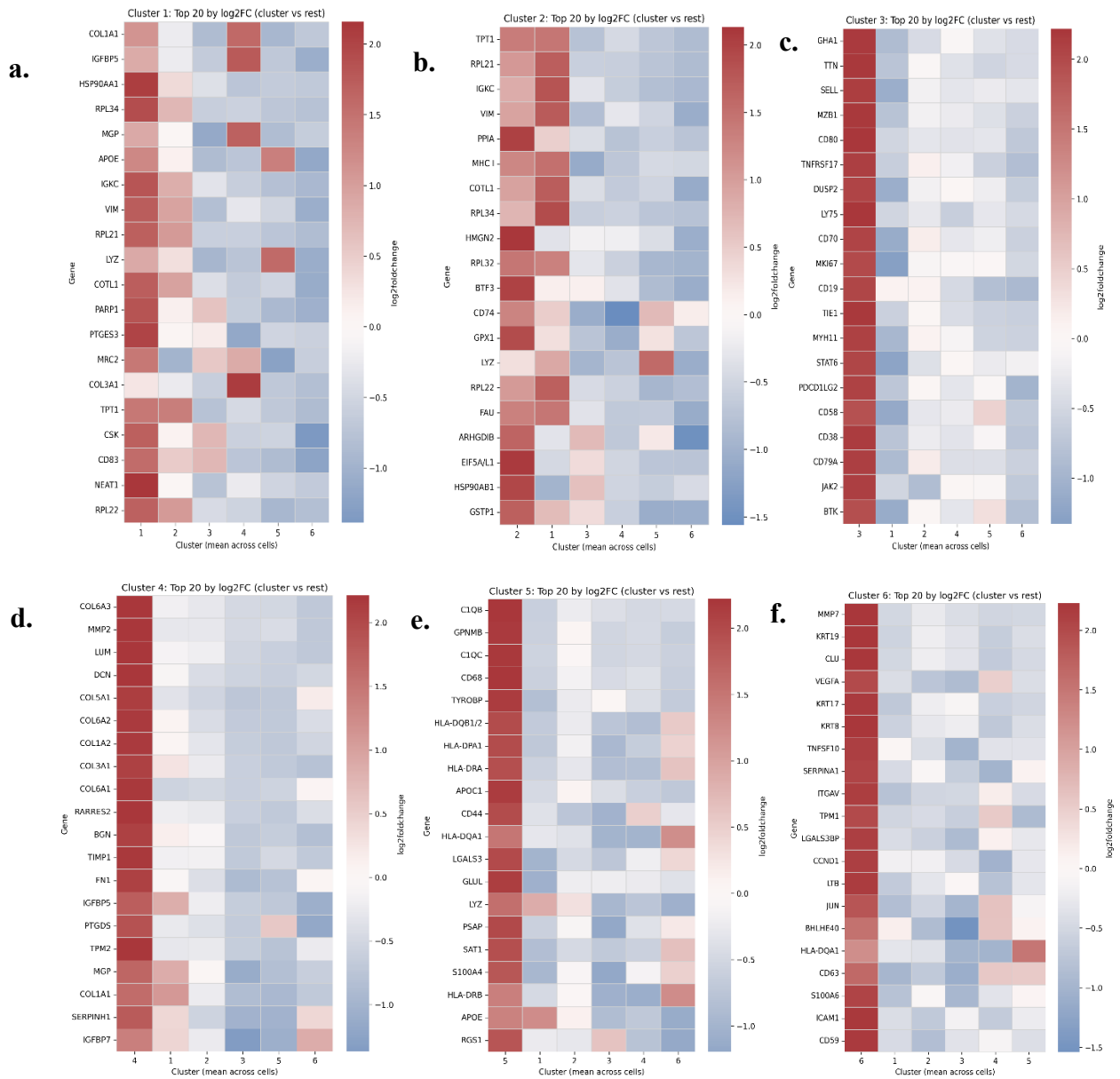


Figure 6: MAST-like Top 20 heatmaps Panels (a–f) show clusters 1–6. Rows are the Top 20 = FDR-significant genes with the largest positive log2FC in the target cluster (BH-adjusted FDR<0.05); columns are clusters. The colour scale displays cluster-wise mean log2FC (target vs rest).

Cell-type annotation and spatial validation

Cluster markers were compiled as the union of the Top 20 genes per cluster from three approaches (Wilcoxon with FDR control, a MAST-like hurdle model, and Random-Forest feature importance). These markers were compared against curated reference catalogues, and a label was assigned when two or more sources converged on the same lineage with clear marker matches, or when a dominant lineage-specific signature was supported by canonical genes. Using this procedure, clusters were annotated on the UMAP. A post-hoc spatial audit across FOVs 103–106 then confirmed coherent tissue localisation for these groups and identified Cluster 1 as a margin-localised, housekeeping-biased set consistent with edge capture artefact; Cluster 1 is therefore reported for completeness but excluded from biological interpretation.

Annotated UMAP with Cluster Labels

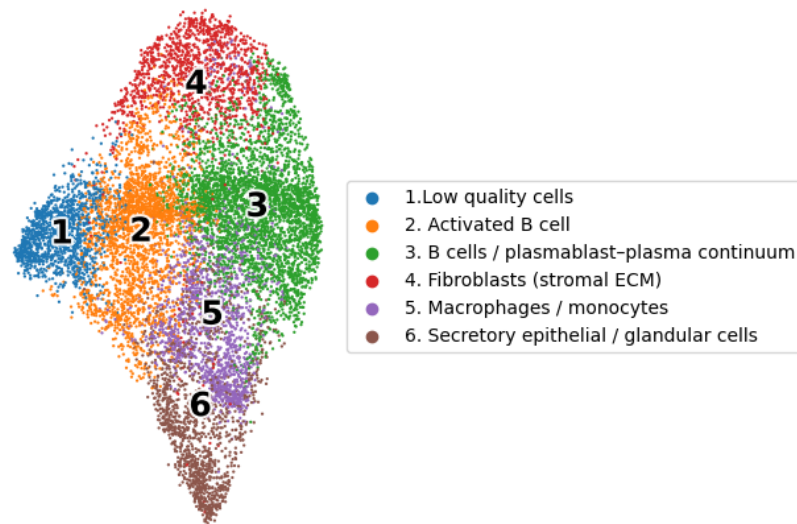
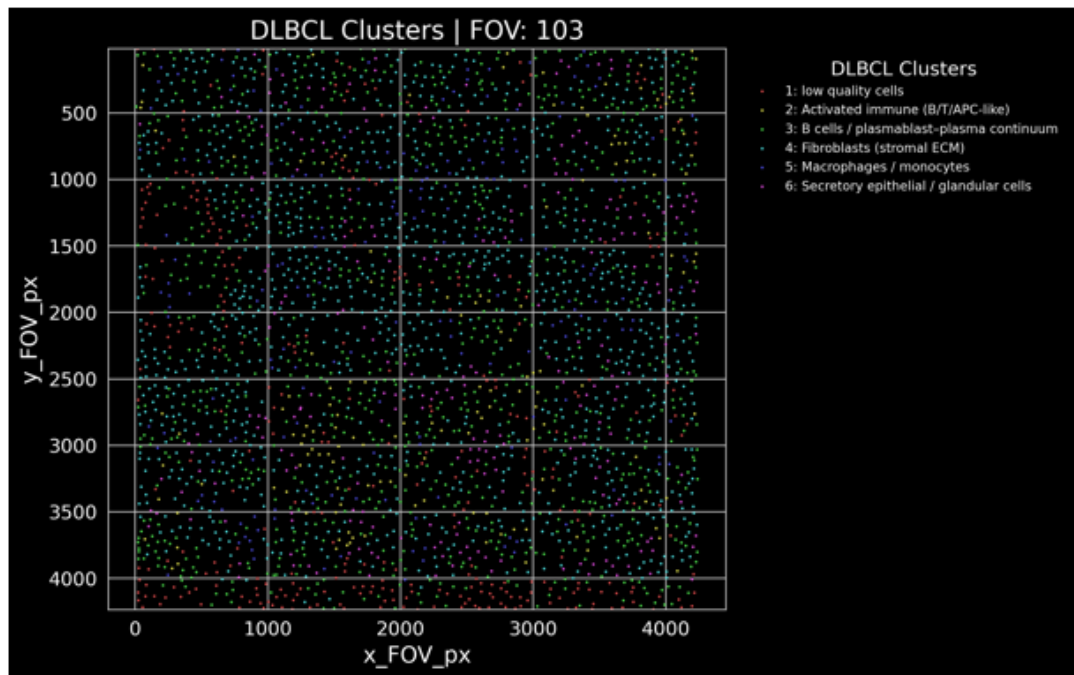


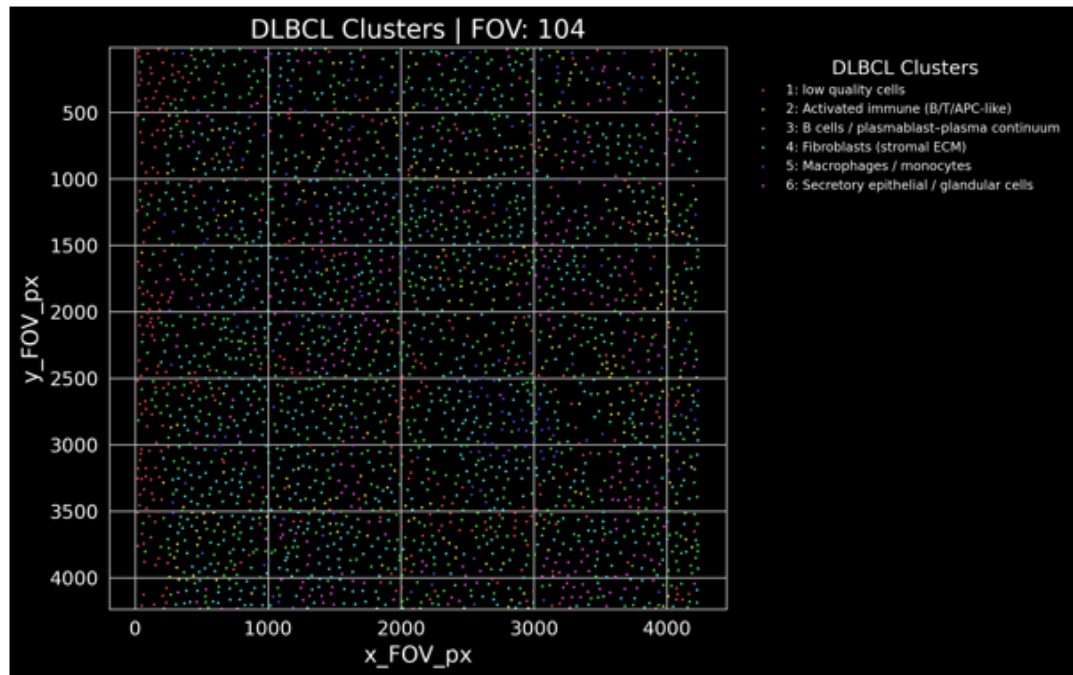
Figure 7: UMAP embedding was generated from the 12-principal-component representation of the full 989-gene panel after quality control (11,382 cells). Clusters were defined by Ward's hierarchical clustering in the PCA space and cut at $k = 6$. Each point corresponds to a single cell; colours and overlaid numerals denote cluster identity: **1**, low-quality/edge cells; **2**,

activated B-cell/immune; **3**, B cells/plasmablast–plasma continuum; **4**, fibroblasts (stromal ECM); **5**, macrophages/monocytes; **6**, secretory epithelial/glandular cells. Cluster 1 is displayed for completeness but excluded from downstream biological interpretation, as its members localised predominantly to tissue edges and showed low-content expression patterns confirmed in subsequent spatial overlays.

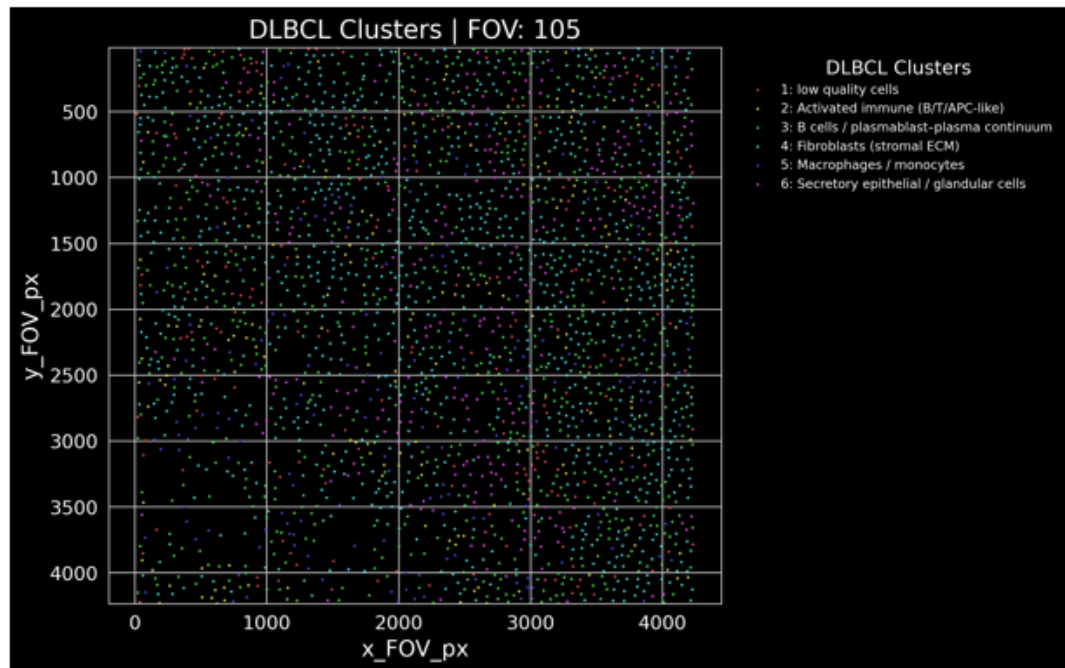
a.)



b.)



c.)



d.)

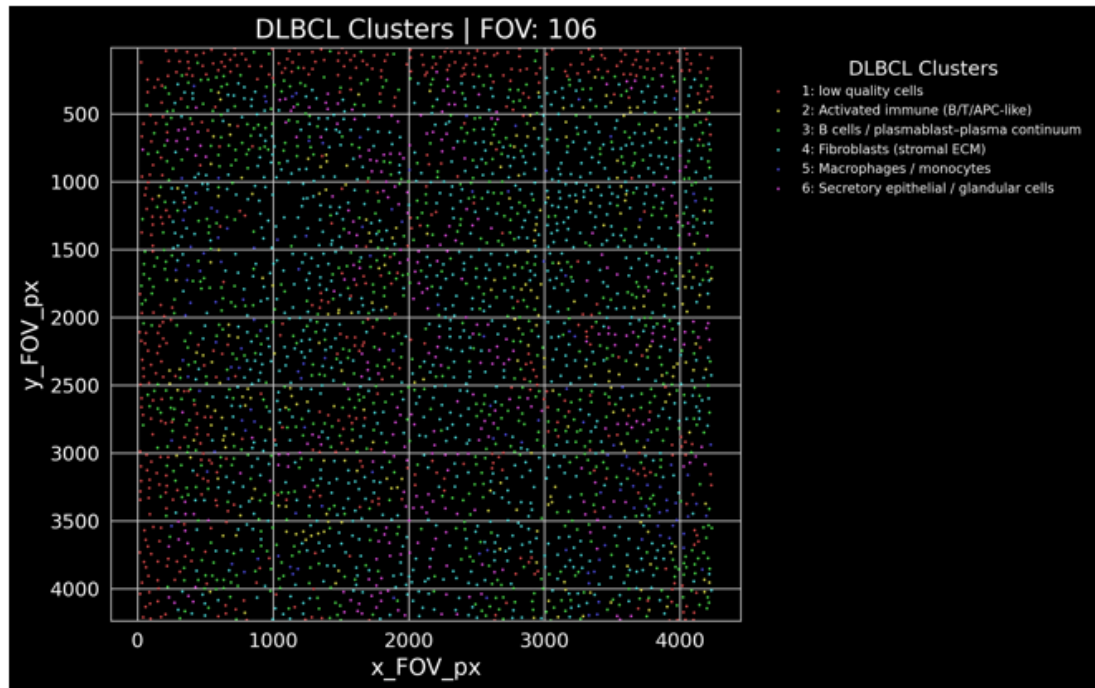


Figure 8. Per-cell cluster labels are plotted in CosMx image coordinates (x_FOV_px , y_FOV_px); colours match the annotated UMAP (Cluster 1 = low-quality/edge; 2 = activated B-cell/immune; 3 = B cells/plasmablast–plasma continuum; 4 = fibroblasts/ECM; 5 = macrophages/monocytes; 6 = epithelial). (a) FOV 103: Cluster 1 cells concentrate along the inferior (bottom) margin. (b) FOV 104: Cluster 1 cells enrich along the left (lateral) edge. (c) FOV 105: Cluster 1 cells form a central band across the FOV, consistent with an internal tissue boundary or low-capture corridor (e.g., fold/lumen/tear), rather than biological tissue; other clusters populate the surrounding parenchyma. (d) FOV 106: Cluster 1 cells localise along the left edge. Across FOVs, Cluster 1 repeatedly occupies non-parenchymal zones (outer edges or internal voids), supporting its designation as a technical/low-quality group, whereas clusters 2–6 show coherent tissue organisation.

Cluster Number	Wilcoxon	Random forest	Mast	Finalized Annotations	Justification
2	Macrophage Mesenchymal cell Activated B-cells	Plasma cell Mesenchymal cell Plasmablast	Plasma cell Plasma Cell Activated B cells	Activated B cell	Mixed immune activation markers (<i>TPT1</i> , <i>IFITM1</i> , <i>IGKC</i> , <i>CD74</i>) suggest an activated B/T/APC population.
3	B cell Memory B cell Plasma cells	B cell Memory B cell Plasma cells	Plasma cell Plasmablast B cell plasmablasts	B cells / plasmablast– plasma continuum	Canonical B-cell lineage markers (<i>CD19</i> , <i>MS4A1</i> , <i>CD79A</i>) combined with plasma differentiation genes (<i>IRF4</i> , <i>MZB1</i> , <i>JCHAIN</i>) confirm B/plasma identity.
4	Fibroblast Fibroblast Fibroblast	Fibroblast Fibroblast fibroblasts	Fibroblast Fibroblast Fibroblasts	Fibroblasts (stromal ECM)	Abundant extracellular matrix genes (<i>COL1A1</i> , <i>COL3A1</i> , <i>FNI</i> , <i>LUM</i> , <i>DCN</i>) are characteristic of fibroblasts
5	Macrophage Pan-macrophage Macrophages	Microglial cell Pan-macrophage Macrophages	Microglial cell Pan-macrophage macrophages	Macrophages / monocytes	Myeloid lineage markers (<i>LYZ</i> , <i>CD68</i> , <i>TYROBP</i> , <i>APOE</i> , <i>C1QA/B/C</i>) define macrophage/monocyte identity.
6	Epithelial cell Immature B cell Epithelial cells	Epithelial cell Immature B cell Luminal epithelial cells	Epithelial cell Epithelial cell Activated epithelial cells	epithelial cells	Cytokeratins (<i>KRT19</i> , <i>KRT8</i>) together with secretory/injury markers (<i>MMP7</i> , <i>SERPINA1</i> , <i>CLU</i> , <i>VEGFA</i>) point to epithelial cell identity.

Table 2. DLBCL cluster annotations with method-specific Top 20 summaries and external reference support. Each row corresponds to one cluster. For Wilcoxon, Random Forest, and MAST-like columns, we first computed the Top 20 marker genes per cluster using the named method, then queried curated catalogues to obtain the dominant matching cell types for that Top 20 set. Multiple entries within a cell indicate the most frequent/consistent matches returned across catalogues. The Finalised annotations column records the label chosen after adjudication; the Justification column provides a one-line rationale using canonical markers. Wilcoxon and the MAST-like model were run with FDR control (full thresholds in Methods); Random Forest supplies feature-importance rankings (no *p*-values) and serves as complementary evidence. Catalogue colour key used within the table: ACT (red), CellMarker 2.0 (green), CellKb (purple), CZ-CellxGene Gene Expression (orange). A label was accepted when ≥ 2 catalogues converged on the same lineage with clear marker matches, or when a

dominant lineage-specific signature was corroborated by canonical genes.

Note: Cluster 1 (low-quality/edge artefact) is not included in this table and is excluded from biological interpretation; see spatial figures for evidence.

Discussion

The data were analysed on log-normalised, log_{1p}-transformed expression values without additional scaling, which preserved effect sizes while stabilising variance a pragmatic approach for targeted CosMx panels where gene abundances differ by biology rather than technical bias. Marker discovery was triangulated with three complementary approaches. Wilcoxon provides a robust non-parametric comparison tolerant of sparse, zero-inflated counts³⁹. MAST's hurdle model explicitly separates detection from expression level, which suits spatial data with variable capture efficiency. Random forest adds an orthogonal perspective by identifying multigene, non-linear features that best discriminate each cluster. Taking the top twenty genes from each method and then checking for agreement yielded stable signatures for clusters two to six, while the absence of three-way overlap in cluster one signalled poor specificity.

Cluster definition and the choice of cluster number were guided by indices that make minimal assumptions about geometry. Although the silhouette (−1 to 1; higher better) was modest at 0.163, this is expected when clusters are unequal in size, elongated, or connected by transitional phenotypes patterns typical of lymphoid tissues where activation states bridge compartments³². In contrast, the Calinski–Harabasz index (higher better) emphasises between-cluster separation relative to within-cluster dispersion, while the Davies–Bouldin index (lower better) penalises within-cluster scatter relative to separation. The favourable CH (2384.693) and DB (1.654) therefore provided complementary evidence that the Ward

partition at $k = 6$ captured meaningful structure, a conclusion reinforced by downstream biological coherence of the clusters^{30,31,40}. The six-cluster solution also mapped to interpretable biology: activated B cells, a B cell to plasmablast/plasma continuum, fibroblasts with extracellular-matrix programmes, macrophage–monocyte populations with antigen-presentation signatures, and a secretory epithelial compartment.

Validation came both from spatial overlays and from prior knowledge. Canonical markers recovered by the three methods aligned with established descriptions of the DLBCL microenvironment collagens and matrix modulators in fibroblasts, complement and HLA class II in macrophages, cytokeratins and secretory genes in epithelium, and B-lineage markers reflecting plasmablast–plasma differentiation in malignant cells. The residual cluster localised to tissue borders and low-capture corridors, expressed mainly housekeeping-biased genes, and was excluded from biological interpretation.

Taken together, variance-stabilised log-normalised data, geometry-agnostic clustering metrics, and convergent evidence from Wilcoxon, MAST, and random forest provide a consistent account of DLBCL heterogeneity that is concordant with both the published literature and the spatial organisation visible in CosMx images.

Our study has several limitations. First, it analyses a single diagnostic DLBCL specimen (four FOVs) using a targeted 989-gene CosMx panel, which constrains transcriptome breadth and generalisability; spatial coordinates were deliberately not used, so spatial relationships were only checked post hoc and not modelled in clustering. Quality control thresholds (library size $\leq 1,000$; $n_genes \leq 350$; mitochondrial $< 15\%$) and the $\geq 25\%$ within-cluster detection rule were chosen to suppress low-content profiles but may also exclude rare yet biologically relevant states. Methodologically, PCA dimensionality (12 PCs), Euclidean distance with Ward linkage, and selection of $k=6$ by internal indices (CH/DB; modest

silhouette) could bias towards approximately spherical, well-separated groups and understate continuous phenotypes typical of lymphoid tissues. Differential analysis combined Wilcoxon, MAST and random forest; while complementary, these approaches are sensitive to zero inflation, correlated predictors and class imbalance, and random-forest importances can be unstable in correlated gene sets. Cell-type labels relied on catalogue matches and the CZ CellxGene portal rather than orthogonal validation (e.g., copy-number inference, BCR clonality or protein markers), so malignant versus reactive B-cell boundaries remain provisional. Finally, we did not correct for potential FOV-specific effects, doublets or segmentation artefacts, and we lacked an independent cohort for external validation; together, these factors warrant cautious interpretation and motivate replication across additional tissues and alternative clustering/annotation strategies. Top of FormBottom of Form

Conclusion

In summary, hierarchical clustering combined with complementary marker discovery approaches yielded a coherent, biologically sensible map of the tissue. By applying three differential expression methods and cross-checking the top 20 markers against ACT, CellMarker 2.0, and CellKb, we finalised labels with the CZ Cellxgene portal and saw clear agreement for activated B cells, a B cell–to–plasmablast/plasma continuum, fibroblasts with extracellular matrix programmes, macrophages/monocytes, and epithelial cells. Cross-method overlaps were strong for most clusters, supporting robust annotation, while Cluster 1 stood out with no three-way overlap. Inspection of all FOVs showed Cluster 1 cells concentrated at tissue edges with low content, indicating a heterogeneous mixture that likely included technical or edge artefacts. This explains the lack of shared markers across Wilcoxon, MAST, and random forest for Cluster 1 and justifies excluding it from biological interpretation. Overall, the combined statistical and spatial evidence supports confident cell-type

assignments for clusters 2–6 and provides a reliable foundation for downstream biological insight.

Future work

In future analyses, the pipeline will be extended to additional DLBCL tissues and other samples, with all parameters re-optimised to reflect each dataset's unique distributions and variance structure. Particular focus will be placed on testing variations of hierarchical clustering, including alternative distance metrics (e.g. Euclidean, correlation, cosine) and linkage strategies (Ward, average, complete), alongside comparisons to graph-based approaches such as Leiden and Louvain. This will allow assessment of how parameter choices influence cluster structure and biological interpretability. In parallel, systematic exploration of volcano plots will be undertaken to better understand how fold-change and significance interact across methods, especially for sparse or zero-inflated genes. Comparing results across samples, tissues, and clustering strategies will help establish the robustness of marker signatures and strengthen confidence in the biological conclusions.

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